

In the context of development economics, the Human Development Index (HDI) incorporates which of the following conceptual improvements over GDP per capita as a measure of development?

- A) It adjusts GDP for purchasing power parity only
- B) It incorporates freedom and democratic participation
- C) It combines health, education, and income dimensions into a geometric mean
- D) It measures environmental sustainability as a fourth dimension

Answer: C

How does the New Institutional Economics perspective redefine the development process compared to traditional neoclassical approaches?

- A) By focusing solely on human capital formation
- B) By emphasizing transaction costs, property rights, and governance structures
- C) By treating development as an automatic outcome of market liberalization
- D) By discounting the role of formal institutions in favor of cultural factors

Answer: B

Which of the following is an implication of the Lewis model of structural change in relation to labor markets in developing countries?

- A) Wages in the urban sector adjust quickly to equilibrium
- B) Migration from rural to urban sectors stops once capital accumulation ceases
- C) Urban wages are fixed above marginal product of labor in rural areas, incentivizing labor transfer
- D) Labor productivity is uniformly distributed across sectors

Answer: C

In the endogenous growth models (e.g., Romer model), long-run economic growth is primarily sustained by:

- A) Capital deepening and exogenous technology
- B) Accumulation of human capital and knowledge spillovers
- C) Trade liberalization under perfect competition
- D) Government budget surpluses and fiscal discipline

Answer: B

Which of the following is the main assumption limiting the application of the Solow growth model to many developing countries?

- A) Labor is perfectly substitutable with capital
- B) Technological progress is endogenous
- C) Capital markets are perfectly functioning
- D) Technological change is exogenous and universally accessible

Answer: D

One key controversy surrounding import-substitution industrialization (ISI) policies is that:

- A) They successfully created globally competitive industries
- B) They increased foreign exchange reserves in developing countries
- C) They led to inefficient domestic monopolies behind high tariff walls
- D) They were compatible with WTO rules and free trade norms

Answer: C

Which of the following best captures the criticism of using only the Gini coefficient in assessing inequality?

- A) It fails to distinguish between vertical and horizontal inequality
- B) It is sensitive to income transfers among the very poor
- C) It assigns higher weight to inequality in high-income groups
- D) It ignores differences in household composition and size

Answer: A

Which policy best combines pro-growth and pro-equity objectives in development planning?

- A) Trade liberalization without safety nets
- B) Land redistribution accompanied by access to credit and education
- C) Fiscal tightening and inflation targeting
- D) Removal of all price controls in agricultural markets

Answer: B

How does the concept of the 'dependency burden' affect economic development in low-income countries?

- A) It reduces consumption levels across generations
- B) It creates incentives for fertility reduction

- C) It strains public services and reduces capital accumulation per worker
- D) It reflects income inequality among the working-age population

Answer: C

Which of the following best explains the concept of 'education externalities' in the context of human capital theory?

- A) Private benefits of education always exceed social returns
- B) The benefits of education accrue only to the educated individual
- C) Educated individuals can enhance productivity and reduce crime in their communities
- D) Education systems do not require public investment

Answer: C

The 'brain drain' phenomenon impacts developing countries by:

- A) Enhancing domestic innovation systems
- B) Improving remittance flows without trade-offs
- C) Depleting the stock of skilled workers and weakening institutions
- D) Providing direct foreign investment opportunities

Answer: C

What does the concept of 'urban bias' imply for development strategy?

- A) Governments prioritize rural areas due to food security concerns
- B) Rural infrastructure is overfunded relative to urban centers
- C) Policy distortions favor urban elites at the expense of rural development
- D) Migration is discouraged by providing urban subsidies

Answer: C

Which of the following explains the phenomenon of 'urban-rural terms of trade' and its impact on agricultural development?

- A) Rising rural prices incentivize food production
- B) Government-imposed low agricultural prices transfer surplus to urban consumers and industries
- C) Agricultural protectionism distorts export competitiveness
- D) Urban prices fall relative to rural costs

Answer: B

Why is land reform considered a crucial institutional intervention in developing agrarian economies?

- A) It ensures monoculture development
- B) It increases the capital-labor ratio in agriculture
- C) It improves equity, productivity, and access to credit for smallholders
- D) It supports large-scale mechanization

Answer: C

What was one major flaw in import-substitution industrialization (ISI) strategies widely adopted in the 1960s–1970s?

- A) Excessive dependence on global markets
- B) Lack of emphasis on domestic consumption
- C) Inefficient allocation of resources due to lack of competition
- D) Overinvestment in rural infrastructure

Answer: C

Which of the following criticisms is most associated with the 'aid dependency' hypothesis?

- A) Aid increases domestic tax revenue
- B) Aid strengthens long-term fiscal institutions
- C) Aid discourages domestic resource mobilization and accountability
- D) Aid enhances political stability in all cases

Answer: C

What is the primary objective of the Heavily Indebted Poor Countries (HIPC) Initiative?

- A) Increase short-term foreign direct investment flows
- B) Encourage privatization of public enterprises
- C) Provide debt relief to ensure debt sustainability and promote poverty reduction
- D) Expand military expenditures through foreign borrowing

Answer: C

What is true about a random or stochastic process?

- A) It is a collection of deterministic variables
- B) It is a collection of random variables ordered in time
- C) It is a collection of variables with no specific order
- D) It is a collection of variables that are not related each other

Answer: B

If $Y_t = b_0 + b_1 Y_{t-1} + b_2 X_t + b_3 X_{t-1} + U_t$; then, X_{t-1} , are lagged exogenous variables, while Y_{t-1} is

- A) lagged exogenous variables
- B) lagged endogenous variable.
- C) current exogenous variables
- D) current endogenous variable

Answer: B

Whereas the LPM assumes that P_i is linearly related to X_i , the logit model assumes

- A) The log of the odds ratio is linearly related to X_i .
- B) The antilog of the odds ratio is linearly related to X_i .
- C) The odds ratio is linearly related to X_i .
- D) The logit model is linearly related to X_i .

Answer: A

In simultaneous equation models, how is the number of equations determined?

- A) Equal to the number of explanatory variables
- B) Equal to the number of time periods
- C) Equal to the number of jointly dependent variables
- D) Equal to the number of error terms

Answer: C

To ease parameter estimation, the logit model can be linearized by taking its natural log, then, the log of the odds ratio is

- A) linear only in variable X
- B) linear only in coefficient β s
- C) linear both in X and the β s
- D) linear in only intercept coefficient

Answer: C

What is a Random Walk without a drift?

- A) A stochastic process with a constant term
- B) A deterministic process with no constant term
- C) A deterministic process with a constant term
- D) A stochastic process with no constant term

Answer: D

Which of the following is a common type of panel data model?

- A) Fixed Effects Model

- B) Random Effects Model
- C) Dynamic Panel Data Model
- D) All of the above

Answer: D

Let's assume the Keynesian model given by:

$$C = \alpha + bY + U$$

$Y = C + Z$. whereas C = consumption expenditure; Z = non-consumption expenditure, Y = national income. Which of the following is exogenous?

- A) Y
- B) C
- C) Z
- D) B

Answer: C

Which of the following statements is true regarding the dummy variable trap?

- A) occurs when there is no variation in the dependent variable
- B) occurs when there is perfect collinearity or perfect multicollinearity among the independent variables
- C) occurs when there is a perfect correlation between the dependent variable and one of the independent variables
- D) Occurs when all categories of a qualitative variable are introduced as dummy variables in a regression model
- E) B and D

Answer: E

What is the primary advantage of using panel data over pure cross-sectional or time-series data?

- A) It allows for the analysis of behavior across time and entities simultaneously
- B) It reduces the number of observations required for analysis
- C) It eliminates the need for statistical testing
- D) It simplifies data collection processes

Answer: A

Suppose that in Nigeria and South Korea, the output per worker for producing rice and ships is shown in the table below.

Country	Rice (in tons)	Ships (in units)
<i>Nigeria</i>	<i>80</i>	<i>2</i>
<i>S. Korea</i>	<i>100</i>	<i>15</i>

Based on this information, which of the following statements is false?

- A) South Korea has an absolute disadvantage in the production of ships.
- B) The domestic opportunity cost ratio is 40:1 for Nigeria.

- C) A given input in South Korea is 7.5 times as productive as its counterpart in Nigeria in shipbuilding.
- D) The domestic opportunity cost ratio in South Korea is 6.67:1.

Answer: A

For two trading nations (Nation I and Nation II), which of the following would cause a deterioration in Nation I's terms of trade?

- A) Negative change in the test pattern of Nation II against Nation I's products
- B) Technological innovation in Nation I
- C) Increase in demand for Nation I's products in Nation II
- D) Deterioration of Nation I's terms of trade

Answer: A

Which of the following best describes the assumption of 'no factor intensity reversal' in the Heckscher-Ohlin theorem?

- A) A product is produced using different techniques in different countries.
- B) Two commodities have the same factor intensity within a given country.
- C) The production functions for a good are identical across all countries.
- D) The same factor is used in the same proportion to produce a good across all countries.

Answer: C

Which of the following statements is not true according to the Overlapping Demand Theory?

- A) Trade between rich and poor nations is unlikely to exist
- B) Countries with similar per-capita income tend to have similar tastes
- C) Western countries are most likely to trade with each other
- D) Demand-side factors largely explain trade in manufactured goods

Answer: A

In the Imitation and Demand Lag Hypothesis, international trade occurs only when:

- A) Demand lag exceeds imitation lag
- B) Imitation lag exceeds demand lag
- C) Demand lag is greater than the innovation lag
- D) Demand lag is at least as long as imitation lag

Answer: B

The idea that a country should not depend on foreign markets for its essential needs reflects the:

- A) Self-sufficiency argument
- B) Defense argument
- C) Balance of Payments (BOP) argument
- D) Economic stability argument

Answer: A

The New Trade Theory differs from the Heckscher-Ohlin Model primarily because it allows for:

- A) Workers in one country to be more productive than in another

- B) Costs to decrease as an industry employs more of all factors of production (economies of scale)
- C) Firms to be perfectly competitive in all markets
- D) Countries to export different goods than their trading partners

Answer: B

The Factor Price Equalization theorem in the Heckscher-Ohlin Model suggests that:

- A) Free trade causes the income of identical factors in different countries to become more similar
- B) If a country does not trade, skilled and unskilled workers will earn the same wages
- C) Trade equalizes the returns to human and physical capitals across countries
- D) All workers in different countries will have identical productivity

Answer: A

Based on the Factor Price Equalization theorem, if Ethiopia (labor-abundant) and the USA (capital-abundant) trade freely, then:

- A) Wages will increase in the USA and decrease in Ethiopia
- B) The price of capital will rise in the USA and fall in Ethiopia
- C) Wages will increase in Ethiopia and decline in the USA
- D) Both A and B

Answer: C

The correct order from the lowest to highest degree of regional economic integration is:

- A) Customs Union, Economic Union, Common Market, Free Trade Area
- B) Free Trade Area, Customs Union, Economic Union, Common Market
- C) Free Trade Area, Customs Union, Common Market, Economic Union
- D) Free Trade Area, Economic Union, Customs Union, Common Market

Answer: C

Which organization is primarily responsible for providing short-term financial assistance and policy advice to countries facing balance-of-payments problems or economic crises?

- A) International Monetary Fund (IMF)
- B) World Health Organization (WHO)
- C) United Nations Educational, Scientific and Cultural Organization (UNESCO)
- D) World Bank (WB)

Answer: A

What will be the potential action for Ethiopia under a fixed exchange rate system to counter a weakening Birr?

- A) Decrease its gold reserves
- B) Sell its foreign currency reserves
- C) Increase domestic inflation
- D) Peg the Birr to a new currency

Answer: B

From the following, which one is an example of hedging in the foreign exchange market is?

- A) Using a forward contract to reduce risk
- B) Taking a long position in a foreign currency
- C) Speculating on future exchange rate movements
- D) Arbitraging between two markets

Answer: A

The long-term exchange rate movements are primarily determined by:

- A) Trade barriers
- B) National price levels
- C) Short-term interest rates
- D) Government interventions

Answer: B

According to the Absorption Approach, what determines whether devaluation improves the Balance of Payments?

- A) The change in national income relative to domestic absorption.
- B) The level of foreign reserves.
- C) The exchange rate regime.
- D) The interest rate differential between countries.

Answer: A

According to the Mundell-Fleming Model, what happens to a small open economy with perfect capital mobility under fixed exchange rates if it attempts monetary expansion?

- A) Output increases, and the interest rate falls.

- B) Output remains unchanged due to capital outflows and reserve losses.
- C) The exchange rate depreciates, improving the current account.
- D) Fiscal policy becomes ineffective.

Answer: B

Which policy combination is most effective for achieving both internal and external balance under fixed exchange rates?

- A) Fiscal expansion and monetary contraction.
- B) Fiscal and monetary expansion.
- C) Exchange rate devaluation alone.
- D) Sterilized intervention in foreign exchange markets.

Answer: A

What does the official settlements balance represent?

- A) The sum of visible and invisible trade balances.
- B) Operations undertaken by monetary authorities to finance imbalances.
- C) The combined trade and investment balance.
- D) Total inflows and outflows of private sector funds.

Answer: B

Why did the Gold Exchange Standard collapse in 1931?

- A) France's decision to convert pounds into gold
- B) The rise of economic nationalism
- C) The Great Depression of 1929
- D) All of the above

Answer: D

What is the key feature of the Bretton Woods System?

- A) Currencies were pegged to the United States (U.S.) dollar, which was convertible to gold.
- B) Exchange rates were determined purely by market forces.
- C) It eliminated the role of the IMF in global financial stability.
- D) It allowed unrestricted capital flows without fixed exchange rates.

Answer: A

Why did the Bretton Woods System fail to benefit poor countries' (balance of payment) BOP deficits?

- A) Exports of poor nations were agricultural in nature
- B) IMF loans were sufficient but poorly managed
- C) Fixed exchange rates ensured balanced trade
- D) Rich countries refused to support poor countries

Answer: A

The Accelerator Principle suggests that investment is primarily influenced by:

- A) The overall price level.
- B) Changes in the level of income.
- C) The rate of inflation.
- D) Government debt levels.

Answer: B

If a firm anticipates a significant increase in future demand for its products, it is likely to:

- A) Reduce its current investment.
- B) Increase its current investment.
- C) Maintain its current investment without changes.
- D) Decrease its labor force

Answer: B

What is the primary objective of monetary policy?

- A) To directly control the level of government spending.
- B) To set all consumer prices to a fixed level.
- C) To influence the availability and cost of money and credit in the economy.
- D) To manage the supply of goods and services.

Answer: C

Open market operations involve the central bank:

- A) Changing the tax rates for individuals.
- B) Providing direct loans to small businesses.
- C) Buying or selling government securities in the financial markets.
- D) Regulating international trade tariffs.

Answer: C

Which of the following is a potential limitation of monetary policy in stimulating an economy during a severe recession?

- A) It can only affect the supply side of the economy.
- B) Interest rates may already be near zero, limiting further effectiveness ("liquidity trap").
- C) It is always effective regardless of how severe the economic downturn is.
- D) It only impacts the level of government consumption.

Answer: B

An increase in government spending would cause the IS curve to:

- A) Shift to the right.
- B) Shift to the left.
- C) Become flatter.
- D) Become steeper.

Answer: A

A decrease in the money supply would cause the LM curve to:

- A) Shift to the right.
- B) Shift to the left.
- C) Become flatter.
- D) Become steeper.

Answer: B

If an economy is experiencing a deep recession and the central bank implements an expansionary monetary policy, what would be the likely impact on the IS-LM model?

- A) LM curve shifts left, leading to higher interest rates and lower output.
- B) Both IS and LM curves shift to the left.
- C) IS curve shifts right, leading to higher interest rates and higher output.
- D) LM curve shifts right, leading to lower interest rates and higher output.

Answer: D

According to the Keynesian perspective, what is a primary cause of economic fluctuations?

- A) Efficient and self-correcting markets.
- B) Fluctuations in aggregate demand, particularly due to changes in investment and consumption.
- C) Stable and predictable government spending.
- D) Supply-side shocks only, like natural disasters.

Answer: B

If the government increases its budget deficit by borrowing more, what is the most likely direct impact on the loanable funds market?

- A) A decrease in the supply of loanable funds.
- B) An increase in the supply of loanable funds.

- C) A decrease in the demand for loanable funds.
- D) An increase in the demand for loanable funds.

Answer: D

Which macroeconomic school emphasizes the self-correcting nature of markets through price and wage flexibility, and views government intervention as unnecessary?

- A. Keynesian
- B. Classical
- C. Neo-classical
- D. Monetarist

Answer: B

In the IS-LM model, a decrease in government spending would shift:

- A) The IS curve leftward, reducing output and interest rates
- B) The LM curve leftward, increasing interest rates
- C) The IS curve rightward, increasing output
- D) The LM curve rightward, reducing interest rates

Answer: A

Under a fixed exchange rate regime with perfect capital mobility, fiscal policy is:

- A) Ineffective due to crowding out
- B) Highly effective in increasing output
- C) Neutralized by monetary policy
- D) Only effective if paired with tariffs

Answer: B

The Keynesian aggregate supply curve is horizontal in the short run because:

- A) Prices and wages are perfectly flexible
- B) Output is determined solely by demand
- C) Workers suffer from money illusion
- D) Prices and wages are sticky

Answer: D

If nominal GDP is \$1,200 billion and the GDP deflator is 120, what is real GDP?

- A) \$1,000 billion
- B) \$1,200 billion
- C) \$1,440 billion
- D) \$1,080 billion

Answer: A

If investment becomes highly sensitive to interest rates, the IS curve will:

- A) Become steeper
- B) Become flatter
- C) Shift to the right
- D) Shift to the left

Answer: B

In the Mundell-Fleming model, which policy is most effective in increasing output under fixed exchange rates?

- A) Monetary policy
- B) Fiscal policy
- C) Trade quotas
- D) None of the above

Answer: B

What is the derivative of $f(x) = 3x^3 + 2x^2 + x$?

- A) $9x^2 + 4x + 1$
- B) $6x + 2$
- C) $3x^2 + 2x$
- D) $x^2 + 2x + 3$

Answer: A

If $P = 60 - 4Q$, what is the shape of the total revenue function?

- A) Linear
- B) Concave
- C) Convex
- D) Undefined

Answer: B

Newton's method is used for:

- A) Solving integrals
- B) Optimizing cost functions
- C) Approximating roots of equations
- D) Finding elasticity

Answer: C

The first-order condition for a maximum or minimum is:

- A) $f'(x) = 0$
- B) $f(x) = 0$
- C) $f'(x) = 0$
- D) $f(x) > 0$

Answer: C

Which variable is optimized in an objective function?

- A) Constant
- B) Choice variable
- C) Marginal variable
- D) Coefficient

Answer: B

Which condition indicates a maximum in constrained optimization?

- A) Positive definiteness of bordered Hessian
- B) All principal minors being negative
- C) Alternating signs of principal minors
- D) Zero second derivatives

Answer: C

Comparative statics is primarily concerned with:

- A) Equilibrium shifts due to parameter changes
- B) Optimization without constraints
- C) Utility maximization only
- D) Elasticity computation

Answer: A

Jacobian determinants are used to:

- A) Evaluate economic forecasts
- B) Solve systems of linear equations
- C) Determine dependency between functions
- D) Analyze convexity

Answer: C

What is the order and degree of the differential equation $((dy/dx)^3 = x^2)$?

- A) Order = 1, Degree = 1
- B) Order = 3, Degree = 1
- C) Order = 2, Degree = 3
- D) Order = 1, Degree = 3

Answer: D

What is the general solution of the homogenous first-order linear differential equation $dy/dt + ay = 0$?

- A) $y(t) = A + e^{(at)}$
- B) $y(t) = Ae^{(-at)}$
- C) $y(t) = e^{(-at)} + A$
- D) $y(t) = Ae^{(at)}$

Answer: B

According to welfare economics, market failure implies:

- A) Excess demand in a market.
- B) Conditions where markets fail to produce Pareto efficient outcomes due to some action external to the market mechanism
- C) Markets that reach equilibrium efficiently.
- D) The natural functioning of capitalism economic system.

Answer: B

The concept of *deadweight loss* is associated the following except:

- A) Inefficiency caused by monopoly practices
- B) Overall losses aroused from market failure factors that deviates market from equilibrium
- C) Welfare from trade in a perfectly competitive market
- D) Indicate benefits that could have secured had monopolists operate at minimum of AC

Answer: C

Suppose a product's market mechanism fails due to some negative externality. Which government intervention is recommended to correct the market failure?

- A) Determining quotas for each producer
- B) Minimizing suppliers cost through Subsidies
- C) Controlling the market price
- D) Applying Pigovian taxes

Answer: D

What is the concept of *game theory* in the context of oligopoly?

- A) Setting price based on their own product feature
- B) An exemplary of strategic interaction between firms
- C) model of perfect rivalry among the firms
- D) A model of both negative and positive externalities

Answer: B

In market failure theory what is referred by the "tragedy of the commons" ?

- A) Market failure caused by public goods
- B) Market failure caused by common resource depletion
- C) Market failure caused by negative externalities
- D) the situation where market mechanism fails to result Pareto efficiency

Answer: B

Which of the following statements is true regarding the demand curve faced by a firm in monopolistic competition?

- A) Like perfectly competitive firm it faces perfectly elastic demand curve
- B) Unlike perfectly competitive firm it faces perfectly elastic demand curve
- C) Like oligopoly firms, it faces downward sloping demand curve
- D) Unlike perfectly competitive firm, it faces is vertical demand curve

Answer: C

A situation where loss incurred by one of the players is necessarily greater than the gain received by the other player is said to be:

- A) Positive– sum – game
- B) strategic game
- C) negative – sum – game
- D) zero – sum – game

Answer: C

In game theory, a situation in which each player chooses an optimal strategy, given the strategy chosen by the other player is said to be:

- A) Partial equilibrium
- B) General equilibrium
- C) Pay off strategy
- D) Nash equilibrium

Answer: D

A form of price leadership in which firms formally or informally agreed to follow the firm which is considered to have a *good knowledge of the prevailing conditions and can forecast better the future market situation* is termed called:

- A) Barometric price leadership
- B) Natural oligopoly price leadership
- C) Dominant firm price leadership
- D) Low cost price leadership

Answer: A

Which one of the following is not effect if a decrease in the price of labor (wage) when several factor of productions are used under perfect competition market?

- A) The substitution effect increases the employment but reduces capital.
- B) It shifts the MPPL and VMPL curves inwards or to the left.
- C) Output effect increases the use of both labor and capital
- D) Profit maximizing level of the inputs combination will be shrinking inward.

Answer: D

All can be the possible consequence(s) when a firm owns monopoly power in product and monopsony power in factor market *except*:

- A) It pays the factor a price that is less than not only VMP_L but also its MRP_L .
- B) It gives rise to monopsonistic exploitation in addition to monopolistic exploitation.
- C) It pays the factor a price that is less than VMP_L only.
- D) Inefficiency arised from it would be greater than the case of a monopolist in product market only.

Answer: C

Which one of the following statements deemed to be wrong regarding bilateral monopoly?

- A) Bilateral monopoly arises when a single seller (monopolist) faces a single buyer (monopsonist).
- B) The outcome of wage bargaining can be known with certainty.
- C) A bilateral monopoly solution is 'indeterminate' as the model gives only the upper and lower limits.
- D) Wage rate will be determined by bargaining.

Answer: B

A monopsonist's wage rate is typically _____ than the wage rate in a perfectly competitive labor market.

- A) Higher.
- B) Lower.
- C) Equal.
- D) More volatile.

Answer: B

A deadweight loss associated with monopoly arises because:

- A) The monopolist's costs are higher than in perfect competition.
- B) The monopolist produces less output than the socially optimal level.
- C) Consumers have perfectly inelastic demand.
- D) The monopolist always earns positive economic profits.

Answer: B

The value of the marginal product (VMP) of a factor is equal to:

- A) Marginal product multiplied by the quantity of output.
- B) Marginal product divided by the price of output.
- C) Marginal product multiplied by the price of output.
- D) Total revenue divided by the quantity of the factor.

Answer: C

In the Prisoner's Dilemma, the Nash equilibrium outcome is:

- A. Pareto efficient.
- B. Pareto inefficient, with both players choosing to confess.
- C. The outcome where both players remain silent.
- D. Unpredictable.

Answer: B

A subgame perfect Nash equilibrium (SPNE) is a refinement of Nash equilibrium that applies to:

- A) Simultaneous-move games.
- B) Sequential-move games.
- C) Games with perfect information.
- D) Both (b) and (c).

Answer: B

A mixed strategy involves:

- A) Choosing an action with certainty.

- B) Randomizing over two or more possible actions.
- C) Cooperating with the opponent.
- D) Always playing the dominant strategy.

Answer: B

A monopoly is characterized by:

- A) Many sellers selling identical products.
- B) A single seller with significant market power.
- C) Few sellers with differentiated products.
- D) Free entry and exit in the market.

Answer: B

The Law of Diminishing Marginal Returns states that as more of a variable input is added to a fixed input:

- A) Total product will eventually decrease.
- B) Marginal product will eventually decrease.
- C) Average product will eventually decrease.
- D) All of the above.

Answer: B

The supply curve of labor to an individual firm in a perfectly competitive labor market is:

- A) Downward sloping.
- B) Upward sloping.
- C) Perfectly elastic (horizontal).
- D) Perfectly inelastic (vertical).

Answer: C

The kinked demand curve model in oligopoly suggests that a firm will face:

- A) A very elastic demand curve if it raises its price and a very inelastic demand curve if it lowers its price.
- B) A very inelastic demand curve if it raises its price and a very elastic demand curve if it lowers its price.
- C) A demand curve with constant elasticity.
- D) A perfectly elastic demand curve.

Answer: A

The reaction function in a Cournot model shows:

- A) A firm's profit as a function of its own output.
- B) A firm's optimal output level given the output level of its rival.
- C) The market demand curve.
- D) The marginal cost curve.

Answer: B

In a Bertrand model of oligopoly, firms compete by choosing:

- A) Quantity.
- B) Price.
- C) Advertising.

D) Product innovation.

Answer: B

Limit pricing is a strategy where an incumbent firm:

- A) Charges the monopoly price to maximize short-run profits.
- B) Sets a low price to deter entry of new firms.
- C) Matches the price of new entrants.
- D) Raises prices after new firms enter.

First-degree (perfect) price discrimination allows the monopolist to:

- A) Charge different prices based on the quantity consumed.
- B) Charge each customer his or her maximum willingness to pay.
- C) Charge different prices in different markets.
- D) Charge a price equal to marginal cost.

Answer: B

Which of the following best mitigates adverse selection and moral hazard?

- A) High interest rates
- B) Financial intermediaries
- C) Insurance premiums
- D) Tax reforms

Answer: B

To address a liquidity shortage, the central bank lowers the discount rate, leading to a spike in discount window borrowing. What is the direct effect of this borrowing on the monetary base?

- A) It decreases
- B) It increases
- C) It stays constant
- D) Only currency changes

Answer: B

Hana wants a safe place to keep her savings. She does not need to write checks and prefers higher interest. Her brother, Mahdi, needs access to funds regularly and writes many checks. Hana is best suited for a:

- A) Checkable deposit
- B) Time deposit
- C) Reserve deposit
- D) Demand deposit

Answer: B

In the Ethiopian financial system, government bonds are sold to large investors with short maturity. On the other hand, banks invest in long-term corporate loans. The market for short-term bonds is a:

- A) Capital market
- B) Secondary market
- C) Money market
- D) Commodity market

Answer: C

A bank is evaluating how much capital to hold, ensuring it can meet deposit withdrawals and reduce risk of default when lending. Ensuring the bank has sufficient funds to meet withdrawal needs is part of:

- A) Asset management
- B) Liquidity management
- C) Liability control
- D) Discount operations

Answer: B

The National Bank of Ethiopia is committed to both low inflation and full employment. A supply shock raises prices and unemployment simultaneously. What is the central bank's policy dilemma?

- A) Lower inflation while raising employment
- B) Increase inflation to boost employment
- C) Conflicting goals that cannot be achieved simultaneously
- D) No dilemma since both goals are aligned

Answer: C

A bank issues loans to various sectors. To manage its credit risk, it uses customer screening, collateral, and loan performance monitoring. What problem is the bank addressing through these tools?

- A) Overcapitalization
- B) Monetary contraction
- C) Credit risk (default)
- D) Legal risk

Answer: C

An individual investor sells Birr 50,000 in bonds to the National Bank and opts to cash the check received instead of depositing it. They hold the money in physical currency. How will this transaction affect reserves and the monetary base?

- A) Both reserves and monetary base increase
- B) Reserves remain unchanged; monetary base increases
- C) Reserves increase; monetary base remains unchanged
- D) Both reserves and monetary base remain unchanged

Answer: B

Assume that Nib International Bank receives a Birr 200 million discount loan from the National Bank during a liquidity crunch. What is the impact on the monetary base?

- A) It increases by Birr 200 million
- B) It decreases by Birr 200 million
- C) It remains unchanged
- D) It leads to a fall in checkable deposits

Answer: A

Let say Commercial Bank of Ethiopia holds Birr 50 million in excess reserves but chooses not to lend due to risk aversion amid economic uncertainty. What impact does this have on the deposit expansion process?

- A) It continues unaffected
- B) It accelerates
- C) It halts entirely
- D) It depends on currency preference

Answer: C

One of the following is not correct about model

- A) Structural equations express the endogenous variables as a function of other endogenous variables, predetermined variables, and disturbances.
- B) The reduced form of a structural model is a model in which the endogenous variables are expressed as a function of the predetermined variables only.
- C) An econometrics model is used for prediction of the value of dependent variable if and only if the chosen model refutes the hypothesis of economic theory under consideration.
- D) The structural parameters and reduced-form parameters measure different effects.

Answer: C

Which one of the following is incorrect?

- A) Unbiased estimator with the most variance is known as an efficient estimator.
- B) Sampling errors are inevitable in all estimates.
- C) Relationship between X and Y is said to be stochastic if for a particular value of X there is a whole probabilistic distribution of values of Y.
- D) The probability distribution of the OLS estimates $\hat{a}$, $\hat{b}$ and δ^2 can be easily derived.

Answer: A

If the amount of change in one variable is accompanied by the same amount of change in the other variable, it is known as _____

- A) Partial Correlation
- B) Multiple correlation
- C) Linear correlation

D) Positive correlation

Answer: C

A pair of random variables is uncorrelated. In other words, which of the following is true of these variables?

- A) The random variables will have a zero covariance.
- B) Their variances are zero.
- C) Their variances are the same.
- D) The random variables are statistically independent.

Answer: A

What does the null hypothesis (H_0) represent?

- A) A statement suggesting a significant effect or relationship.
- B) A statement asserting no effect or relationship exists.
- C) A claim that the sample mean equals zero.
- D) A prediction about the future values of variables.

Answer: B

What does a p-value represent in hypothesis testing?

- A) The probability that the null hypothesis is true.
- B) The probability of observing the test statistic under the null hypothesis.
- C) The probability of making a Type II error.
- D) The critical value for rejecting the null hypothesis.

Answer: B

Form hypothetical data a weekly family consumption expenditure (Y) and a weekly family income (x) $n=10$.

$$\begin{array}{llll} \bar{X} = 170 & \bar{Y} = 111 & \sum ei^2 = 337.2682 & \sum xi^2 = 33,000 \\ \sum Xi^2 = 322,000 & \sum xiyi = 16,800 & \sum Yi^2 = 8890 & \end{array}$$

Determine the error variance

- A. 0.03
- B. 42.1585
- C. 33.7268
- D. 24.45

Answer: B

Suppose a friend of yours has estimated a simple linear regression function for 30 observations and obtained the following regression line and intermediate results.

$$\hat{Y} = -6.5 + 1.3 X_i$$

$$\sum \hat{e}_i^2 = 133 \quad \sum x_i^2 = 500 \quad \sum y_i^2 = 978$$

Where, the values of $\sum x_i^2$ and $\sum y_i^2$ are the sum of squares deviations from their mean. Given the above information which of the following is wrong?

- A) The coefficient of determination is 0.8640 and $F^* = 177.8945$.
- B) The deviated value of $\sum x_i y_i = 650$
- C) The variance of the estimated coefficient $\hat{\beta}_1$; that is $\text{var}(\hat{\beta}_1) = 0.0095$
- D) The coefficient of correlation is 0.9295 and $t^* = 13.3378$.
- E) None of the above

Answer: E

Suppose a regression model suffers from multicollinearity. Which of the following is a consequence of multicollinearity?

- A) The coefficients are biased.
- B) The variances of coefficients are very large.
- C) The t-statistics are very high.
- D) The coefficients are not consistent

Answer: B

Which method provides the Best Linear Unbiased Estimators (BLUE) when heteroscedasticity is present?

- A) Ordinary Least Squares (OLS)
- B) Weighted Least Squares (WLS)
- C) Generalized Least Squares (GLS)
- D) Instrumental Variables (IV)

Answer: B

The idea of need based project mainly arise from:

- A) Profitable use of available resources
- B) Identified demand in home or overseas market
- C) Local leaders
- D) Need of community to make available some basic materials requirements

Answer: D

The project cycle or stage in which the funds are actually disbursed to start the project and keep running.

- A) project preparation
- B) project appraisal
- C) Implementation
- D) Ex-post evaluation

Answer: C

Which one of the following is non-discounting project selection criteria?

- A) Accounting rate of return
- B) Net present value
- C) Internal rate of return
- D) Profitability index

Answer: A

An asset costs \$150,000 with a \$50,000 salvage value at the end of its five year life. If annual cash inflows are \$50,000, the cash payback period is:

- A) 4 years
- B) 3 years
- C) 6 years
- D) 5 years

Answer: B

Which one of the following is the extra rate that makes an individual indifferent between the receipts at different time periods?

- A) Domestic resource cost ratio
- B) Net benefit investment ratio
- C) Benefit cost ratio
- D) Marginal rate of time preference

Answer: D

A good is tradable good if:

- A. CIF prices > Domestic market prices
- B. FOB prices < Domestic market prices
- C. CIF prices < Domestic market prices
- D. FOB prices = Domestic market prices

Answer: C

A set of prices that are believed to better reflect the opportunity cost or the cost in their best use.

- A) Shadow prices
- B) Border prices
- C) CIF prices

D) FOB prices

Answer: A

A project analyst can apply the parameter directly to the project under analysis. Which one is it?

- A) Conversion Factor
- B) National parameter
- C) Standard Conversion Factor
- D) Social Cost Benefit Analysis

Answer: B

Work breakdown structure is the one of the process of project planning. Which of the following is not true about work breakdown structure?

- A) To make planning effective by dividing works into manageable element which can be easily planned, budgeted and controlled.
- B) To assign responsibility for work elements to project personnel and outside agencies
- C) To develop monitoring, control and information system
- D) To define the work to be done not in detailed manner

Answer: D

Which one of the following is the challenge in the monitoring and evaluation project?

- A) Adequate resources
- B) Standardized indicators
- C) Lack of stakeholder engagement
- D) Sufficient attention to context

Answer: C